## 1.5 Solution sets of linear systems

Solution sets of linear systems are important objects of study in linear algebra. In this section, we use vector notation to give explicit and geometric description of such solution sets.

Homogeneous Linear Systems



- A system of linear equations is said to be homogeneous if it can be written in the form $Ax = 0$ where $A$ is $m \times n$ matrix and 0 is the zero vector in $\mathbb{R}^m$.
- Such a system $Ax = 0$ always has at least one solution, namely, $x = 0$ (the zero vector in $\mathbb{R}^n$) which is called the trivial solution.
- For a given equation $Ax = 0$, the important question is whether there exists a nontrivial solution, that is, a nonzero vector $x$ that satisfies $Ax = 0$.
- Remember the Existence and Uniqueness Theorem from section 2:

**Existence and Uniqueness Theorem**

A linear system is consistent if and only if the rightmost column of the augmented matrix is *not* a pivot column—that is, if and only if an echelon form of the augmented matrix has *no* row of the form

$$[\,0 \quad \cdots \quad 0 \quad b\,] \qquad \text{with } b \text{ nonzero}$$

If a linear system is consistent, then the solution set contains either (i) a unique solution, when there are no free variables, or (ii) infinitely many solutions, when there is at least one free variable.

- This theorem leads immediately to the following fact: The homogeneous equation $Ax = 0$ has a nontrivial solution if and only if the equation has at least one free variable.

Ex: Determine if the following system has a nontrivial solution. Then describe the solution set

$$\begin{aligned} 3x_1 + 5x_2 - 4x_3 &= 0 \\ -3x_1 - 2x_2 + 4x_3 &= 0 \\ 6x_1 + x_2 - 8x_3 &= 0 \end{aligned} \rightarrow \begin{bmatrix} 3 & 5 & -4 & 0 \\ -3 & -2 & 4 & 0 \\ 6 & 1 & -8 & 0 \end{bmatrix}$$

$$\rightarrow \begin{matrix} R_1+R_2 \rightarrow \\ R_3-2R_1 \end{matrix} \begin{bmatrix} 3 & 5 & -4 & 0 \\ 0 & 3 & 0 & 0 \\ 0 & -9 & 0 & 0 \end{bmatrix} \rightarrow R_3+3R_2 \begin{bmatrix} 3 & 5 & -4 & 0 \\ 0 & 3 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & -\frac{4}{3} & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

skipping calculations for reduced echelon form.

basic, basic, free

$x_3$ is free. Therefore $Ax=0$ has nontrivial solutions. (one for each choice of $x_3$).

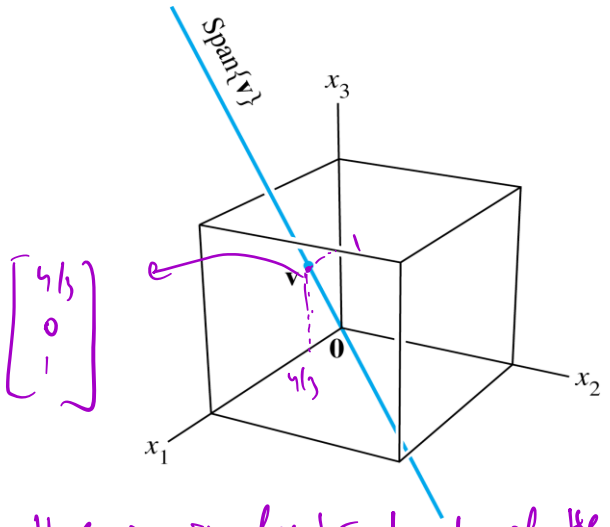
$$\begin{aligned} x_1 - \frac{4}{3}x_3 &= 0 \\ x_2 &= 0 \\ 0 &= 0 \end{aligned}$$

$x_1 = \frac{4}{3}x_3$, $x_2 = 0$, $x_3$ is free

As vector, the general solution of $Ax=0$ has the form:

$$x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} \frac{4}{3}x_3 \\ 0 \\ x_3 \end{bmatrix} = x_3 \begin{bmatrix} 4/3 \\ 0 \\ 1 \end{bmatrix} = x_3 \cdot v$$

where $v = \begin{bmatrix} 4/3 \\ 0 \\ 1 \end{bmatrix}$




$\begin{bmatrix} 4/3 \\ 0 \\ 1 \end{bmatrix}$

Here $x_3$ is factored out of the expression for the general solution vector. This shows that every solution of $Ax=0$, in this case, is a scalar multiple of $v$. Geometrically, the solution set is a line through 0 in $\mathbb{R}^3$.

Ex: A single linear equation can be treated as a very simple system of equations. Describe all the solutions of the homogeneous "system":

$$10x_1 - 3x_2 - 2x_3 = 0$$

No need for matrix notation. The general solution is

$x_1 = 0.3x_2 + 0.2x_3$ with $x_2$ and $x_3$ is free.

$$\begin{aligned} 10x_1 - 3x_2 - 2x_3 &= 0 \\ 0 &= 0 \\ 0 &= 0 \end{aligned} \rightarrow \begin{bmatrix} 10 & -3 & -2 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \rightarrow$$

basic, free, free

As a vector, the general solution is

$$x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0.3x_2 + 0.2x_3 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0.3x_2 \\ x_2 \\ 0 \end{bmatrix} + \begin{bmatrix} 0.2x_3 \\ 0 \\ x_3 \end{bmatrix} = x_2 \begin{bmatrix} 0.3 \\ 1 \\ 0 \end{bmatrix} + x_3 \begin{bmatrix} 0.2 \\ 0 \\ 1 \end{bmatrix}$$

$= x_2 \cdot u + x_3 v$

The solution set is $\text{Span}\{u, v\}$

Above examples illustrate the fact that the solution set of a homogeneous equation $Ax = 0$ can always be expressed as $\text{Span}\{v_1, \dots, v_p\}$ for suitable vectors $v_1, \dots, v_p$.

That equation is called a parametric vector equation of the plane.

Sometimes such an equation is written as

$$x = su + tv \quad (s, t \text{ in } \mathbb{R})$$

to emphasize that the parameters vary over all real numbers.

If there is only one free variable, the solution set is a line through the origin.

A plane through the origin occurs when there are two or more free variables.

Whenever a solution set is described explicitly with vectors, we say that the solution is in parametric vector form.

$Ax = 0$

$Ax=0$

$\rightarrow$ $Ax=b$

Solutions of nonhomogeneous systems

When a nonhomogeneous linear system has many solutions, the general solution can be written in parametric vector form as one vector plus an arbitrary linear combination of vectors that satisfy the corresponding homogeneous system.

Ex: Describe all solutions of $Ax=b$, where $A=\begin{bmatrix} 3 & 5 & -4 \\ -3 & -2 & 4 \\ 6 & 1 & -8 \end{bmatrix}$ & $b=\begin{bmatrix} 7 \\ -1 \\ -4 \end{bmatrix}$

$$\begin{bmatrix} 3 & 5 & -4 & 7 \\ -3 & -2 & 4 & -1 \\ 6 & 1 & -8 & -4 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 4/3 & -1 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 0 & 0 \end{bmatrix} \rightarrow \begin{array}{rl} x_1 & -\frac{4}{3}x_3 = -1 \\ x_2 & = 2 \\ 0 & = 0 \end{array}$$

Thus, $x_1 = \frac{4}{3}x_3 - 1$, $x_2 = 2$, $x_3$ free.

As a vector, the general solution of $Ax=b$ has the form:

$$x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} -1 + \frac{4}{3}x_3 \\ 2 \\ x_3 \end{bmatrix} = x_3 \begin{bmatrix} 4/3 \\ 0 \\ 1 \end{bmatrix} + \begin{bmatrix} -1 \\ 2 \\ 0 \end{bmatrix}$$

$x_3 \begin{bmatrix} 4/3 \\ 0 \\ 1 \end{bmatrix}$: solution set of $Ax=0$ ✓ — $\begin{bmatrix} -1 \\ 2 \\ 0 \end{bmatrix}$: $p$ — whole expression: solution set of $Ax=b$

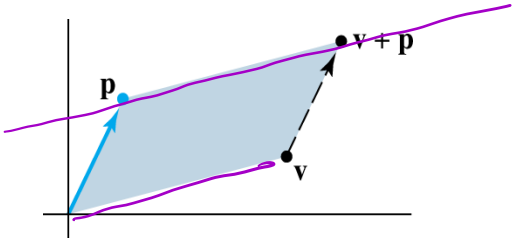
The equation $x = p + x_3 \cdot v$ or writing $t$ as a general parameter

$x = p + t \cdot v$ ($t$ in $\mathbb{R}$)

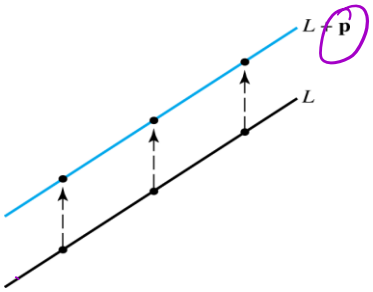
describes the solution set of $Ax=b$ in parametric vector form.

Recall that $Ax=0$ has a solution in parametric vector equation $x = t \cdot v$ ($t$ in $\mathbb{R}$) with the same $v$. Thus, the solutions of $Ax=b$ are obtained by adding the vector $p$ to the solutions of $Ax=0$.

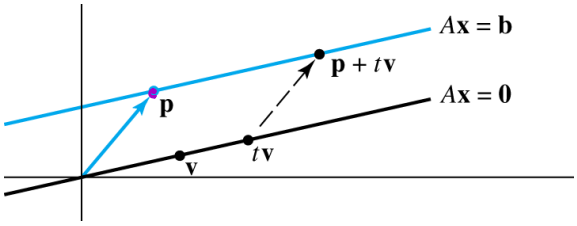
Geometrically adding $p$ to $v$ is to move $v$ in a direction parallel to the line through $p$ and 0. We say that $v$ is translated by $p$ to $v + p$. (See the next figure.)




If each point on a line $L$ in $\mathbb{R}^2$ or $\mathbb{R}^3$ is translated by a vector $p$, the result is a line parallel to $L$. (See the following figure.)




The solutions set of $Ax = b$ is a line through p parallel to the solution set of $Ax = 0$. (See the following figure.)




The relation between the solution sets of $Ax = b$ and $Ax = 0$ shown in the figure above generalizes to any consistent equation $Ax = b$, although the solution set will be larger than a line when there are several free variables.

Ex:

Recall the following example from section 1.2: $Ax=b$

$$\begin{bmatrix} 1 & 6 & 2 & -5 & -2 & -4 \\ 0 & 0 & 2 & -8 & -1 & 3 \\ 0 & 0 & 0 & 0 & 1 & 7 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 6 & 0 & 3 & 0 & 0 \\ 0 & 0 & 1 & -4 & 0 & 5 \\ 0 & 0 & 0 & 0 & 1 & 7 \end{bmatrix} \quad (1)$$

basic, free, basic, free, basic

Given this echelon form, we found the general solution as follows:

(2)

$x_1 = -6x_2 - 3x_4$

$x_2$ is free

$x_3 = 4x_4 + 5$

$x_4$ is free

$x_5 = 7$

Write this general solution in parametric vector form.

$$x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} \overset{(3)}{=} \begin{bmatrix} -6x_2 - 3x_4 \\ x_2 \\ 4x_4 + 5 \\ x_4 \\ 7 \end{bmatrix} = x_2 \begin{bmatrix} -6 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} + x_4 \begin{bmatrix} -3 \\ 0 \\ 4 \\ 1 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 5 \\ 0 \\ 7 \end{bmatrix} \quad (4)$$

$u$ $\quad$ $v$ $\quad$ $p$

$Ax=b$ has a solution set of $s \cdot u + t \cdot v + p$

Then, the solution set of $Ax=0$ is $su+tv$

**Theorem**:

Suppose the equation $Ax = b$ is consistent for some given $b$, and let $p$ be a solution. Then, the solution set of $Ax = b$ is the set of all vectors of the form $w = p + v_h$, where $v_h$ is any solution of the homogeneous equation $Ax = 0$.

Writing a solution set (of a consistent system) in parametric vector form:

1. Row reduce the augmented matrix to reduced echelon form.
2. Express each basic variable in terms of any free variables appearing in an equation.
3. Write a typical solution x as a vector whose entries depend on the free variables, if any.
4. Decompose x into a linear combination of vectors (with numeric entries) using the free variables as parameters.